

A Theoretically Grounded Program for Autocatalytic Set Emergence in PeTTa Algorithmic Chemistry

Design of the rule class, analytic thresholds, attractor basins,
and a sized experimental protocol

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Abstract

This document responds to the PeTTa-Chem external-review packet of 26 July 2026. It argues that the project’s accumulated negative results are, with one or two exceptions, uninformative in a way that can be shown quantitatively rather than by further sweeps, and it proposes a replacement program with three components: (i) a concrete class of rewrite rules with structurally determined catalysis, chosen so that its catalytic density is analytically computable and continuously tunable; (ii) a mean-field theory of the RAF (reflexively autocatalytic, food-generated) closure over that class, which yields a closed-form threshold, predicts a *discontinuous* transition with a nucleation barrier, and identifies production-route multiplicity—not catalytic density—as the binding structural constraint for term rewriting; and (iii) a dynamical theory of seeding and persistence that produces an explicit attractor-basin criterion and a sizing law $NT \gtrsim (Ak_{\text{cat}}\delta^d)^{-1}$ relating population size, horizon, catalytic contrast and bootstrap depth. Applying that law to the existing exp07 configurations shows they were roughly five orders of magnitude below the regime in which spontaneous emergence could have been observed at all. The document closes with a methodological correction—guidance must be used as an unbiased rare-event *estimator*, not as an experimental *treatment*—and an appendix giving a phased, sized, acceptance-criterion-bearing work plan for coding agents.

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1 Purpose and diagnosis

1.1 What the current evidence actually establishes

The review packet is honest and well engineered, and its self-assessment is largely correct. What it does not do is ask whether the experiments were ever *capable* of producing the sought result. In several cases they were not, and this can be shown before running anything further.

exp02 (40 seeds, 8 rules, zero cycles). The standard result in the random catalytic reaction network literature is that RAFs appear once the expected number of reactions catalysed per molecule, f , reaches order unity to a few; the mean-field calculation in Section 4 of this document independently gives $f_c \approx 5\text{--}8$ for systems in the range $|R| \sim 10^2\text{--}10^5$. With $|R| = 8$, achieving $f = 5$ would require each molecule to catalyse 5/8 of *all* reactions in the system. The experiment was never in a regime where the answer could have been anything other than “no cycles”. It carries essentially no evidence about spontaneous emergence.

exp04 (maximal RAF 15, core 2). A greedy core of size 2 inside a 20-rule hand-designed pool is a planted two-rule closure with a large passively food-generated periphery. The shuffle sensitivity reported in the reduction sweep is the correct reading: the RAF is a property of the engineered catalysis map, not of the generative process.

exp07 (any single rule deletion collapses the effect). This is the diagnostic signature of a RAF whose *irreducible core equals the whole set*: there is exactly one irreducible sub-RAF and it has no redundancy. Robust autocatalytic organisation, in every sense that matters downstream, means *many* irreducible sub-RAFTs with overlapping support. The count of irreducible sub-RAFTs should be a standing reported metric; a count of 1 should be treated as *prima facie* evidence of construction rather than emergence.

exp07 “applicability loss”. The mechanism audit found that weak guidance made 51 blocked first-absent selections versus 38 unguided. This is what happens when a selection bias is applied in the *structural* coordinate (which reactions belong to a target closure) while the binding constraint lives in the *dynamical* coordinate (which reactions currently have their reactants present). The report’s own Recommendation 4 identifies this correctly. Section 6 argues the deeper problem is that guidance was used as a treatment at all.

1.2 Three claims

1. **Free term rewriting is structurally hostile to RAFs, for a reason that is fixable in one line of design.** In a free term algebra with a binary constructor, every molecule has exactly one production route (its unique root decomposition). Proposition 1 shows that route multiplicity $r \geq 2$ is *necessary* for any nontrivial closure to exist in mean field. PeTTa-Chem should either make the constructor associative/commutative, work modulo normalisation, or add modification rules supplying alternate routes. Without this, no amount of catalysis or compute will produce a RAF.
2. **The structural transition is discontinuous, with a nucleation barrier.** The mean-field closure map has a saddle-node bifurcation, not a continuous percolation transition (Proposition 3). The closure fraction jumps from 0 to roughly 0.72–0.96 at threshold. This means bistability and hysteresis are intrinsic, and it is the structural shadow of the dynamical basin structure derived in Section 5. It also predicts that the empirical distribution of maximal-RAF sizes should be strongly *bimodal*—a sharp, cheap falsification test.

3. **The binding constraint in the existing experiments was dynamical, not structural, and it was violated by about five orders of magnitude.** Section 5 derives $NT \gtrsim (Ak_{\text{cat}}\delta^d)^{-1}$. The exp07 dual-bootstrap runs had N of order tens of molecules, $T = 32$ ticks, hence $NT \sim 10^3$; with a six-step bootstrap and any honest catalytic contrast this is below the threshold for $d = 3$, let alone $d = 6$. Spontaneous emergence was not observed because it was not observable.

2 Project objective, restated

The scientific claim being pursued is:

There exists a generative distribution over rewrite rules and catalysis relations, defined without reference to any target pathway, such that a stochastic realisation of the induced chemistry, started from food alone, spontaneously develops a reflexively autocatalytic food-generated subsystem that is dynamically active, persistent, causally productive, and robust to deletion of any single element.

Every clause is load-bearing and each is a separate measurement. The rest of this document supplies (a) a candidate distribution, (b) an analytic prediction of where in its parameter space the claim can be true, and (c) a protocol sized to test it.

3 The rule class: pattern-catalysed associative ligation

We want a rule class that is genuinely algorithmic (catalysis is computation, not a random bit), has a conserved quantity (so that dilution and selection are meaningful), has analytically computable catalytic density, and admits a single scalar tuning knob.

3.1 Molecules, mass, and food

Fix an alphabet A of k atoms and a maximum mass M .

Definition 1 (Molecules). *A molecule is a term over a single binary constructor $\cdot$ with leaves in A , taken modulo associativity. Equivalently, a molecule is a nonempty word over A of length at most M . The mass $\mu(s)$ is the word length. The molecule set is $X = \bigcup_{m=1}^M A^m$, with $|X_m| = k^m$ and $|X| = (k^{M+1} - k)/(k - 1)$.*

Definition 2 (Food). *The food set is $F = \{s : \mu(s) \leq m_0\}$ for a fixed $m_0 \geq 1$, held at constant concentration by influx.*

3.2 Reactions and production-route multiplicity

Definition 3 (Reactions). *For every ordered pair (s, t) with $\mu(s) + \mu(t) \leq M$ there is a ligation $\rho_{s,t} : s + t \rightarrow st$ and its reverse cleavage. Mass is conserved by construction.*

The number of ligations producing a specific product of mass m is $m - 1$; the total is

$$|R| = \sum_{m=2}^M (m-1) k^m \sim (M-1)k^M, \quad \frac{|R|}{|X|} \approx M.$$

Definition 4 (Production-route multiplicity). *r_m is the number of distinct reactions in the network whose product is a given molecule of mass m . Here $r_m = m - 1$.*

This is the parameter that free term rewriting gets wrong. In a *free* binary term algebra, a term $u \cdot v$ has a unique root decomposition, so $r_m = 1$ for every m . Associativity restores $r_m = m - 1$. Proposition 1 shows this is the difference between “possible” and “impossible”, not a matter of degree.

- Remark 1** (Three admissible fixes, in order of preference). 1. *Declare $\cdot$ associative and use AC-matching (native and idiomatic in MeTTa). Gives $r_m = m - 1$.*
2. *Work with terms modulo a confluent normalisation, so that many distinct redex reductions land on the same normal form. Gives r_m equal to the number of distinct producing redexes; needs measurement.*
3. *Retain free terms but add unary modification rules $t \rightarrow t'$ that supply alternate routes into each product. Gives r_m equal to 1 plus the in-degree of the modification graph; this is the weakest option because it introduces a second, uncontrolled rule family.*

3.3 Structural catalysis

Catalysis must be determined by the structure of the catalyst, not drawn independently. Otherwise the model is Kauffman’s and the “algorithmic” aspect is decorative.

Definition 5 (Junction-straddling pattern catalysis). *Extend the alphabet with a wildcard symbol $\star$, so molecules are words over $A \cup \{\star\}$. A molecule c is read as a pattern in which $\star$ matches any single atom. Define*

c catalyses $\rho_{s,t} \iff c$ matches, as a pattern, some factor of st that strictly straddles the junction between s and t .

Write $\sigma(c)$ for the specificity of c : the number of non-wildcard positions.

This is biochemically natural (the catalyst binds across the reaction site), it is a genuine computation (pattern matching with wildcards), and its density is computable. For a uniformly random product word, the probability that a fixed pattern matches a fixed alignment is $k^{-\sigma(c)}$, and the number of (reaction, alignment) pairs available to c is $O(Mk^M)$. Hence the expected catalytic out-degree is

$$f(c) = |\{\rho : c \text{ catalyses } \rho\}| \asymp \Theta(M) k^{M-\sigma(c)}. \quad (1)$$

Remark 2 (Pattern catalysis is intrinsically *dense*, which is the opposite of the usual worry). Equation (1) says that to obtain $f = O(1)$ one needs $\sigma(c) \approx M + \log_k M$. But $\sigma(c) \leq \mu(c) \leq M$. Therefore even a maximally specific catalyst—a full-length word with no wildcards—still catalyses $\Theta(M)$ reactions, namely the $M - 1$ ligations that assemble it from its own prefixes and suffixes. In this chemistry $f \gtrsim M - 1$ always. The system is generically supercritical for structural RAF existence.

This is a substantive prediction, and it is consistent with everything in the review packet: static RAFs were found whenever the pool was rich enough, and were never the bottleneck. The bottleneck is dynamical. It also means that to observe a structural transition at all, catalysis must be deliberately diluted.

3.4 The thinning parameter

To sweep across the structural threshold while preserving structural correlations and the determinism the PeTTa harness requires, introduce an explicit thinning:

Definition 6 (Thinned pattern catalysis). *Fix $\theta \in [0, 1]$ and a deterministic hash $h : X \times R \rightarrow [0, 1]$ (seeded per network). Then*

$$c \text{ catalyses } \rho \iff (c \text{ pattern-matches } \rho) \wedge (h(c, \rho) < \theta).$$

This gives a single scalar with $f(\theta) = \theta f_{\max}$, preserves exact replay, preserves the monotonicity property used in Section 4.6, and lets the same code run the iid Kauffman control (replace the pattern-match conjunct by true) at matched f . **The comparison of thinned pattern catalysis against iid catalysis at matched mean degree is the central novel measurement of the whole program:** same f , different correlation structure, and the theory predicts a different threshold.

4 Structural theory: when does a RAF exist?

4.1 RAF and the algorithm as a greatest fixed point

Definition 7 (RAF). *A subset $S \subseteq R$ is a RAF with respect to food F if $S \neq \emptyset$ and, writing $\text{cl}_F(S)$ for the closure of F under the reactions of S : (i) food-generated: every reactant of every $\rho \in S$ lies in $\text{cl}_F(S)$; and (ii) reflexively autocatalytic: every $\rho \in S$ has at least one catalyst in $\text{cl}_F(S)$.*

The RAF algorithm is the iteration $S_{n+1} = \Phi(S_n)$ started from $S_0 = R$, where Φ deletes every reaction that fails (i) or (ii) with respect to $\text{cl}_F(S_n)$. Φ is monotone and decreasing, so the iteration converges to the *greatest fixed point*, which is the unique maximal RAF. If it is empty, no RAF exists. This is the standard polynomial-time algorithm and it is the object the mean-field theory below approximates.

Remark 3. *Existence of a RAF is decidable in polynomial time, and the maximal RAF is unique. Finding a smallest RAF is, to my recollection, NP-hard, and the number of irreducible sub-RAFs can be exponential. Both facts matter operationally: report maxRAF cheaply and always; enumerate irreducible sub-RAFs only for small systems, and report the count as the robustness metric.*

4.2 The mean-field closure recursion

Let q_m denote the probability that a uniformly random molecule of mass m lies in the closure of the maximal RAF. Let $\pi = f/|R|$ be the probability that a given molecule catalyses a given reaction, and let

$$C = \sum_{m=1}^M k^m q_m$$

be the expected size of the closure. The probability that a given reaction has *some* available catalyst is

$$\gamma = 1 - (1 - \pi)^C \approx 1 - e^{-\pi C}. \quad (2)$$

A molecule of mass $m > m_0$ is in the closure iff at least one of its $m - 1$ production routes is live, i.e. both reactants are in the closure and the reaction is catalysed. Under a mean-field independence assumption,

$$q_m = 1 - \prod_{i=1}^{m-1} (1 - q_i q_{m-i} \gamma), \quad q_m = 1 \text{ for } m \leq m_0. \quad (3)$$

Iterating (3) together with (2) from the initial condition $q \equiv 1$ downward converges to the greatest fixed point, mirroring the RAF algorithm exactly. This correspondence is the reason to trust the qualitative conclusions even though the independence assumption is crude.

4.3 Route multiplicity is necessary

Proposition 1 (Unique production routes forbid closure). *Suppose $r_m = 1$ for all $m > m_0$, i.e. every molecule above food mass has exactly one production route. Then (3) reduces to $q_m = q_i q_{m-i} \gamma$ for the unique split $(i, m - i)$, and the only fixed point with q_m bounded is $q_m \rightarrow 0$ geometrically in m . No nontrivial closure exists for any $\gamma \leq 1$.*

Proof. With $r = 1$ the closure map on the scalar summary $\bar{q}$ is $g(\bar{q}) = \gamma \bar{q}^2$. Since $\gamma \leq 1$, $g(\bar{q}) \leq \bar{q}^2 < \bar{q}$ for all $\bar{q} \in (0, 1)$, so g has no fixed point in $(0, 1)$; the greatest fixed point is 0. Level by level, $q_m = \gamma^{m/m_0-1} \prod q_{\text{food}}$ decays geometrically. $\square$

Corollary 2. *A free binary term algebra cannot support a RAF under this dynamics regardless of catalytic density. The observed inability of PeTTa-Chem to produce non-planted closures is, under this model, structurally forced rather than a parameter-tuning failure.*

4.4 The threshold is a saddle-node, not a percolation transition

Consider the scalar reduction of (3) with uniform r routes: $g(q) = 1 - (1 - \gamma q^2)^r$.

Proposition 3 (Discontinuous transition). *$q = 0$ is always a fixed point of g and is always locally stable, since $g'(0) = 0$. Nontrivial fixed points appear in pairs through a saddle-node bifurcation at a critical $\gamma_c(r)$. In the large- r limit, writing $u = r\gamma$ and $g(q) \approx 1 - e^{-uq^2}$, the tangency conditions $g(q) = q$, $g'(q) = 1$ reduce to*

$$\frac{1-v}{2v} = -\ln v, \quad v = 1 - q,$$

with root $v_c = 0.28467$, giving

$$q_c = 0.71533, \quad u_c = \frac{1}{2q_c(1-q_c)} = 2.4554.$$

Numerically solving the finite- r tangency gives:

r	γ_c	$r\gamma_c$	q_c (jump size)
2	0.84375	1.6875	0.8889
3	0.64743	1.9423	0.8260
4	0.51757	2.0703	0.7962
5	0.42944	2.1472	0.7790
7	0.31931	2.2352	0.7599
9	0.25379	2.2841	0.7496
11	0.21048	2.3152	0.7432
13	0.17975	2.3368	0.7388
20	0.11891	2.3783	0.7304
50	0.04849	2.4246	0.7213
100	0.02440	2.4400	0.7183
∞	—	2.4554	0.7153

Three consequences.

1. **$r\gamma \gtrsim 2.4$ is the design target.** With $r = M - 1 = 11$ at $M = 12$, one needs $\gamma \approx 0.21$: about one reaction in five must have an available catalyst. This is a directly measurable quantity and should be instrumented in every run.
2. **Bimodality.** Because the transition is first order, maximal-RAF size should be bimodal across sampled networks—either essentially zero or covering $\gtrsim 70\%$ of the molecule set—with very few intermediate values. A histogram of maxRAF size over 200 sampled networks near threshold is a one-hour falsification test of this entire framework.
3. **The nucleation barrier is intrinsic.** The unstable middle fixed point is a structural barrier that exists before any dynamics is imposed. It is the same object that reappears in Section 5 as the critical seed size. Guidance schemes that push the system across it are doing nucleation, and should be analysed as such.

4.5 Graded predictions for concrete system sizes

Solving the full graded system (2)–(3) numerically for the PCAL chemistry gives the following predicted thresholds. Here f_c is the critical mean catalytic out-degree and γ_c the corresponding catalysed-reaction fraction; q_M is the closure fraction at the top mass just above threshold.

k	M	m_0	$ X $	$ R $	f_c	q_M above
2	6	1	126	516	4.67	0.903
2	8	1	510	3 076	5.70	0.938
2	10	1	2 046	16 388	6.55	0.952
2	12	1	8 190	81 924	7.32	0.959
2	14	1	32 766	393 220	8.04	0.965
2	12	2	8 190	81 924	4.66	0.919
4	8	1	87 380	582 544	5.95	0.926
4	10	1	1 398 100	12 116 880	6.74	0.939

Reading the table.

- f_c grows only *logarithmically* in $|R|$: three decades of system size cost less than a factor of two in required catalysis. This reproduces, from an independent derivation, the qualitative “RAFTs are easy” scaling found in the random catalytic network literature, and it is the single most useful fact for sizing: **you can scale $|R|$ by orders of magnitude without needing proportionally more catalysis.**
- Enlarging the food set from $m_0 = 1$ to $m_0 = 2$ drops f_c from 7.32 to 4.66, a 36% reduction for one extra food level. Food-set size is a stronger lever than catalytic density and is currently untouched in PeTTa-Chem.
- Mean field neglects positive correlations and therefore *overestimates* the threshold (see Section 4.6). The measured f_c should come in at or below these numbers. If it comes in substantially *above*, the catalysis model is hub-dominated (Section 4.7) and the mean degree is a misleading summary.

4.6 A rigorous lower bound via positive correlation

Pattern catalysis has a property iid catalysis lacks: the catalysis events are *monotone* in a partial order. If c pattern-matches a junction of st , then so does any molecule containing c as a factor with compatible wildcards; and adding atoms to a product word can only create, never destroy, matching windows. Concretely, order the randomness by the subword-containment order on the sampled alphabet content. Then:

Proposition 4 (FKG lower bound, schematic). *Let E_ρ be the event “reaction ρ has an available catalyst in the closure” under thinned pattern catalysis with the thinning hash held fixed. The events $\{E_\rho\}$ are increasing functions of a common product measure, hence positively associated by the Harris–FKG inequality. Consequently*

$$\Pr\left[\bigcap_{\rho \in S} E_\rho\right] \geq \prod_{\rho \in S} \Pr[E_\rho],$$

so the probability that a candidate RAF S survives the closure test is at least its independent (mean-field) value. The true threshold satisfies $f_c^{\text{true}} \leq f_c^{\text{MF}}$.

Two remarks. First, this is a genuine advantage of structural catalysis for *theory*: with iid catalysis one is forced into second-moment methods, whereas here the correlations point the

helpful way and give one-line bounds. Second, the proposition as stated is schematic—the precise construction of the underlying product measure needs care, particularly regarding whether the closure itself (which is determined by the same randomness) can be conditioned on without breaking monotonicity. I would treat this as a conjecture worth a short proof effort, not as an established result.

4.7 Degree distributions: hubs versus bulk

Equation (1) gives $f(c) \asymp \Theta(M)k^{M-\sigma(c)}$, which is *exponential in specificity*. If catalyst specificity varies across the molecule pool, the catalytic out-degree distribution is therefore heavy-tailed by construction: a small number of low-specificity, wildcard-rich molecules catalyse almost everything, and the bulk catalyse almost nothing.

This matters because mean-field theory is driven by $f = \mathbb{E}[\text{deg}]$, whereas RAF robustness is driven by the *bulk* of the distribution. A hub-dominated network will show a healthy f , will often contain a maximal RAF, and will have exactly one irreducible sub-RAF that dies when the hub is deleted. That is precisely the exp07 phenotype.

Design rule. Constrain the wildcard content so that catalytic out-degree is *concentrated*—approximately Poisson with mean in the single digits—rather than heavy-tailed. Operationally: sample catalysts with specificity in a narrow band $\sigma \in [\sigma^* - 1, \sigma^* + 1]$, and *report the full out-degree histogram, its Gini coefficient, and its tail index in every experiment*. Mean degree alone should never again be reported as the catalysis parameter.

5 Dynamical theory: reachability, seeding, and attractor basins

Structural RAF existence is a statement about a graph. The scientific claim is about a trajectory. This section supplies the missing dynamical layer and yields the sizing law.

5.1 The stochastic model

Run the chemistry as a continuous-time Markov jump process in a well-stirred chemostat (CSTR):

- state: an integer abundance vector $n \in \mathbb{Z}_{\geq 0}^X$, total population $N = \sum_x n_x$;
- catalysed reaction propensity $k_{\text{cat}} n_s n_t n_c / N^2$;
- uncatalysed propensity $k_0 n_s n_t / N$, with $\varepsilon := k_0 / k_{\text{cat}}$;
- food influx holding n_x for $x \in F$ near a set point;
- dilution: every species removed at rate ϕ , so total population is stationary.

Two parameters carry the entire dynamical story: the *catalytic contrast* ε and the *dilution rate* ϕ .

5.2 Catalytic depth and the seeding rate

Definition 8 (Catalytic depth). *For a target RAF S , the catalytic depth $d(S)$ is the minimum, over all trajectories from the food-only state to a state from which S is self-sustaining, of the number of uncatalysed reaction events on that trajectory.*

d is exactly the length of the shortest bootstrap chain: the number of steps that must occur before any member of S ’s catalyst set is present in copy number. Each such step is suppressed by ε . Hence the rate at which the system nucleates S is

$$\lambda_{\text{seed}} \asymp A N k_{\text{cat}} \varepsilon^d, \quad (4)$$

where A is a combinatorial prefactor counting the number of distinct minimal bootstrap paths (the multiplicity of the transition state, in chemical-physics language). A is exactly what a shortest-path enumeration measures, and it is the one factor that can be made large by design rather than by compute.

Remark 4 (d is a design variable, not a fact of nature). d is set by how far the target closure’s catalysts sit from food. Chemistries whose minimal RAF core lies within one or two ligations of the food set have $d \in \{1, 2\}$ and can nucleate in feasible runs. The exp07 dual-bootstrap fixed a six-step bootstrap. Under (4) that is ε^6 , which for any ε small enough for catalysis to be detectable is astronomically suppressed. Minimising d should be an explicit design constraint on the next chemistry.

5.3 Survival once seeded: the reproductive number

Once a core cycle exists in copy number, treat it as a branching process. For a cycle of length L with per-step conversion efficiency $\eta_i = k_i/(k_i + \phi)$ (the probability that an intermediate is converted before it is diluted) and branching factor A_{br} (the number of copies of the entry species produced per turn),

$$R_0 = A_{\text{br}} \prod_{i=1}^L \eta_i \approx A_{\text{br}} \eta^L, \quad \eta = \frac{k_{\text{cat}}}{k_{\text{cat}} + \phi}. \quad (5)$$

Standard branching-process theory gives extinction probability $R_0^{-n_0}$ from n_0 initial copies, so

$$P_{\text{surv}}(n_0) = 1 - R_0^{-n_0}, \quad R_0 > 1. \quad (6)$$

Corollary 5 (Maximum sustainable cycle length). $R_0 > 1$ requires $L < L_{\text{max}} = \ln A_{\text{br}} / \ln(1/\eta)$. With $\phi = 0.1k_{\text{cat}}$ (so $\eta = 0.909$) and $A_{\text{br}} = 2$, $L_{\text{max}} = 0.693/0.0953 \approx 7.3$. With $\phi = 0.5k_{\text{cat}}$ ($\eta = 0.667$), $L_{\text{max}} \approx 1.7$.

This is a hard constraint that is currently unchecked in PeTTa-Chem, and it interacts badly with the depth constraint: long cycles are both harder to nucleate (ε^d , d grows with L) and harder to sustain (η^L). **Short cores are not a convenience; they are a requirement.**

5.4 The critical seed size and the deterministic basin

Deterministically, the core concentration x obeys, schematically, $\dot{x} = kx(\text{resource}) - \phi x - \beta x^2$, giving an unstable fixed point

$$x^* \approx \frac{\phi}{k[\text{resource}]} \quad (7)$$

below which dilution wins and above which the autocatalytic cycle grows. The basin of attraction of the active ACS is $\{x > x^*\}$; x^* is the dynamical image of the unstable middle fixed point of Proposition 3.

The check that should be run before anything else. In an integer-abundance simulation, if $Nx^* > \max_x n_x$ achievable given the food influx and dilution settings, then *the ACS basin is unreachable regardless of seeding*, and the experiment is a null by construction. Compute Nx^* for the existing exp07 chemistries. My expectation, from the reported abundances and replenishment schedule, is that this alone accounts for a substantial part of the negative results.

5.5 The emergence law

Combining (4), (6):

$$P(\text{emergence and persistence by time } T) = 1 - \exp\left[-A N k_{\text{cat}} \varepsilon^d (1 - R_0^{-n_0}) T\right]. \quad (8)$$

This is a scaling form with four measurable inputs (A, d, R_0, ε) and two control knobs (N, T) . It is falsifiable in the strong sense: sweep N , T , ε , and d independently and check that the measured emergence probabilities *collapse* onto (8). If they do, the project has a quantitative theory of its own subject matter and can size any subsequent experiment analytically. If they do not, the residual identifies which assumption fails (well-stirredness, the single-bottleneck assumption, or the independence of seeding and survival).

5.6 The identifiability window and the sizing law

There is a tension. For catalysis to be *detectable*, the catalysed regime must dominate the uncatalysed one, requiring $\varepsilon \leq \delta$ for some contrast threshold δ (a 20-fold contrast, $\delta = 0.05$, is modest; 100-fold, $\delta = 0.01$, is comfortable). For nucleation to *occur* within the run, (4) requires $ANk_{\text{cat}}\varepsilon^dT \gtrsim 1$. Together:

$$(ANk_{\text{cat}}T)^{-1/d} \lesssim \varepsilon \lesssim \delta, \quad (9)$$

which is nonempty only if

$$NT \gtrsim \frac{1}{A k_{\text{cat}} \delta^d}. \quad (10)$$

required NT (taking $A = k_{\text{cat}} = 1$)			
d	$\delta = 0.1$	$\delta = 0.05$	$\delta = 0.01$
1	10	20	10^2
2	10^2	4×10^2	10^4
3	10^3	8×10^3	10^6
4	10^4	1.6×10^5	10^8
5	10^5	3.2×10^6	10^{10}
6	10^6	6.4×10^7	10^{12}
8	10^8	2.6×10^{10}	10^{16}

5.7 Why exp07 could not have worked

The dual-bootstrap successor used a six-step bootstrap, cap 8, 32 seeds, and ticks 3–34. Taking N as order tens of molecules and $T = 32$ ticks gives $NT \sim 10^3$. Reading off the table at $d = 6$: even at a generous $\delta = 0.1$ one needs $NT \sim 10^6$, and at an honest $\delta = 0.05$ one needs $NT \sim 6 \times 10^7$. The experiment was between three and five orders of magnitude short. The observed 13/32 unguided incidence is therefore not evidence of anything about spontaneous emergence; it is a measurement of how often the constructed pathway happened to be traversed by the deterministic selection rule.

The same table explains why guidance appeared to help once and then failed: guidance was effectively reducing d by fiat in one particular constructed chemistry, and the size of that reduction was an artifact of the pathway design rather than a property of the guidance policy.

6 Measuring rare emergence correctly

6.1 The methodological error

In exp07, Doob- h reweighting was used as an experimental *treatment*: run guided, run unguided, compare incidence. This is the wrong use of the tool, and it is why the mechanism audit ended at “applicability loss”. Any reweighting of the transition kernel changes the process, so the guided arm measures a different chemistry, and no amount of ablation recovers the quantity of interest.

The quantity of interest is $P(\text{spontaneous emergence})$ under the *unmodified* dynamics. When that probability is 10^{-8} , direct simulation cannot measure it and comparison-to-control cannot substitute for measuring it.

6.2 Guidance as an unbiased estimator

Let $\mathbb{P}$ be the true path measure and $\mathbb{Q}$ a guided (tilted) path measure absolutely continuous with respect to it. For any path functional Ψ (e.g. the indicator that a RAF becomes active and persists),

$$\mathbb{E}_{\mathbb{P}}[\Psi] = \mathbb{E}_{\mathbb{Q}}\left[\Psi \frac{d\mathbb{P}}{d\mathbb{Q}}\right], \quad (11)$$

and for a jump process the likelihood ratio is the explicit product over jumps

$$\frac{d\mathbb{P}}{d\mathbb{Q}}(\omega) = \prod_j \frac{a_{\rho_j}(n_{j-})}{\tilde{a}_{\rho_j}(n_{j-})} \exp\left[\int_0^T (\tilde{a}_0(n_s) - a_0(n_s)) ds\right],$$

with $a, \tilde{a}$ the true and tilted propensities and $a_0, \tilde{a}_0$ their totals. Accumulating this weight along each trajectory converts guided sampling into an *unbiased* estimator of the unguided emergence probability, with a variance that is small precisely when the guidance is close to the optimal (Doob h -transform) tilt. This is the correct role for every piece of guidance machinery already built: keep the policies, add the weight accumulator, delete the guided-versus-unguided comparison.

6.3 Splitting methods

Importance sampling with a hand-chosen tilt has heavy-tailed weights in high dimension. The robust alternatives are *splitting* methods, which need only a reaction coordinate, not a tilt:

- **Forward Flux Sampling (FFS).** Define a monotone progress coordinate $\lambda(n)$ —for this problem, the number of distinct target-RAF reactions that have fired at least once, or the size of the largest active sub-RAF—and a ladder $\lambda_0 < \lambda_1 < \dots < \lambda_S$. Estimate $P = \prod_s p_s$ where p_s is the conditional probability of reaching λ_{s+1} before falling back to λ_0 , each measured by firing many short trajectories from stored interface configurations.
- **Adaptive Multilevel Splitting (AMS).** Same idea with adaptively chosen interfaces; generally more robust when the good reaction coordinate is unknown.

Cost. With per-stage crossing probability $p_s \approx 0.2$, reaching $P = 10^{-12}$ takes $S = \log(10^{-12})/\log(0.2) \approx 17$ stages. At 10^3 trial trajectories per stage this is about 1.7×10^4 short trajectory segments—versus the 10^{12} full runs direct simulation would require. This single change buys roughly eight orders of magnitude of reach and is the difference between the sizing table above being prohibitive and being routine.

6.4 What the shortest-path proposal actually computes

The *Let's Get Chemical* proposal—compute shortest paths from a random state to a state with a known ACS, then mine them—is the right instinct, and the theory above says precisely what it yields. In the small-noise limit, the probability of a rare transition is governed by a large-deviation rate function, and the minimum-action path from the food-only state to the ACS state has action

$$I = d \ln(1/\varepsilon) + O(1),$$

because each uncatalysed step contributes $\ln(1/\varepsilon)$. Therefore:

- the **shortest path length in uncatalysed steps** is d , the exponent in the sizing law;

- the **number of distinct minimal paths** is A , the prefactor in (4);
- the **path ensemble is the optimal importance-sampling tilt**, i.e. the empirical approximation to the Doob h -function, obtained without solving any control problem.

So the proposal is not a heuristic shortcut: it directly measures the two quantities that dominate emergence probability, and it supplies the tilt that makes (11) low-variance. The correct pipeline is **shortest paths** $\rightarrow$ **estimate** $(d, A) \rightarrow$ **build tilt** $\rightarrow$ **importance-sample or FFS** $\rightarrow$ **unbiased $P(\text{emergence})$ with error bars**, not **shortest paths** $\rightarrow$ **guidance policy** $\rightarrow$ **compare to control**.

6.5 Exact validation on a tiny state space

Before any of this is trusted, validate on a chemistry small enough to solve exactly: $k = 2$, $M = 4$ ($|X| = 30$, $|R| \approx 40$), bounded total population $N \leq 12$, so the full CTMC has at most a few million states. On this system:

1. compute the exact committor $h(n) = \Pr[\text{reach ACS before returning to food-only} \mid n]$ by sparse linear solve;
2. compute the exact emergence probability by matrix exponential;
3. verify that the FFS and importance-sampling estimators recover it to within their stated error bars;
4. verify that the Doob h -transform built from the exact committor gives zero-variance estimation, as it must.

Only after this should “Doob transform” or “Schrödinger bridge” appear in any write-up as more than an inspiration.

7 Reporting standard

Every run should report four *separate* endpoints, never conflated, plus five diagnostics.

Endpoints.

1. **Structural.** Does the sampled network contain a RAF? Report maxRAF size and the number of irreducible sub-RAFs.
2. **Reachable.** Are the maxRAF’s molecules producible from food under the actual dynamics within the horizon, ignoring persistence?
3. **Active / persistent.** Does a qualifying subset fire and persist for a prespecified fraction of a post-burn-in window?
4. **Causal / productive.** Does deleting its catalysis reduce a declared productivity endpoint relative to a degree-preserving matched null?

A weaker endpoint must never be reported under a stronger label. This is the review packet’s Recommendation 3 and it is correct.

Diagnostics (new, and load-bearing).

1. **Production-route multiplicity** r_m , empirical histogram. If $r_m \equiv 1$, stop.
2. **Catalytic out-degree distribution:** full histogram plus mean, Gini coefficient, tail index. Never the mean alone.
3. **Catalysed-reaction fraction** γ , measured directly. Compare to $\gamma_c \approx 2.4/r$.
4. **Catalytic depth** d and minimal-path multiplicity A of the observed or target core, from shortest-path enumeration.
5. **Estimated R_0 and critical seed size Nx^*** , compared to achievable abundances.

8 Falsification plan

The framework above makes five predictions that can each kill it cheaply:

1. **Route multiplicity.** A free-term-algebra chemistry ($r = 1$) yields zero RAFs at *any* catalytic density; the same chemistry with an associative constructor yields RAFs above threshold. Cost: hours.
2. **Bimodality.** Near threshold, the maxRAF size histogram over sampled networks is bimodal, with a gap covering roughly $0.1 < q_M < 0.6$. Cost: hours.
3. **Threshold location.** Measured f_c lands at or below the graded mean-field predictions (4.7 – 8.0 across $|R| = 5 \times 10^2$ to 4×10^5), and grows logarithmically in $|R|$. Cost: days.
4. **Correlation effect.** At matched mean degree f , thinned pattern catalysis and iid catalysis give *different* thresholds. Direction predicted: structural catalysis has the lower threshold, by FKG. Cost: days. *This is the publishable result.*
5. **Emergence collapse.** Measured emergence probabilities across independent sweeps of N , T , ε , d collapse onto (8). Cost: weeks, and requires the rare-event machinery.

If (1) and (2) fail, the mean-field picture is wrong and this document should be discarded. If (1)–(4) hold, PeTTa-Chem has an analytic instrument and can size every subsequent experiment on paper.

9 Relation to the review packet’s recommendations

The packet’s Recommendations 0 through 5 are broadly right and this document mostly sharpens them.

- **Rec. 0 (repair, stop accretion).** Agreed. The exp03 singleton-answer regression is a bounded fix and should not be a blocker for design work in parallel.
- **Rec. 1 (neutral random chemistry).** Agreed, with the crucial addition that the naive neutral model must have $r_m \geq 2$ and that “neutral” should not mean “iid catalysis” — the whole scientific point is the structured case, with iid as the control.
- **Rec. 2 (validate RAF against an oracle).** Agreed and highest priority. Exhaustive subset enumeration is feasible for $|R| \leq 18$.
- **Rec. 3 (separate four endpoints).** Agreed without modification; add the five diagnostics above.
- **Rec. 4 (applicability-aware guidance).** Superseded. Do not repair guidance as a treatment; convert it into an estimator (Section 6).
- **Rec. 5 (postpone exp08).** Agreed. The 24-permutation design is careful but answers a question about one constructed artifact. Reuse the paired degree-preserving null and exact replay inside the calibration program.

10 A note on references

This document was written without access to search or a bibliographic database. The mathematics above is derived from scratch and the numerical tables were computed directly; those are checkable independently of any citation. However, the surrounding literature claims—the RAF formalism and its polynomial-time algorithm, the NP-hardness of minimum RAF, the empirical “ f of order a few” threshold in the binary polymer model, and the existence of reference RAF implementations—are recalled from memory and **should be verified before being cited**. The relevant bodies of work to check are those of Hordijk and Steel on RAF theory and

algorithms, Mossel and Steel on probability of self-sustaining autocatalysis, Kauffman on the original binary polymer model, Fontana and Buss on algorithmic chemistry and organisations, Vasas, Szathmáry and Santos on the limits of compositional heredity, and Allen and Frenkel or Rolland and Simonnet on forward flux sampling and adaptive multilevel splitting. Author–year pairs and exact constants from these sources should not be reproduced from this document without checking.

A Work plan for coding agents

Six phases. Each has explicit deliverables, explicit sizes, and an acceptance criterion that can be evaluated without judgement. **No phase begins before its predecessor’s acceptance criterion is met.** Predictions from Phase 2 must be committed before Phase 3 data exists.

How to read the time estimates. Each phase heading quotes two independent quantities. *Implementation* is agent wall-clock time spent writing, testing and validating code; it does not parallelise across cores and is the dominant cost of this program. *Compute* is measured in **single-core CPU-hours** on a compiled host, per the throughput assumptions in Appendix B.5. Every compute workload here is embarrassingly parallel at the replicate or grid-cell level with no shared state, so wall time divides by core count essentially exactly: the 50–200 CPU-hours of Phase 6 is under six hours on a 32-core workstation. The two quantities should never be added.

Phase 0 — Repair, freeze, and two cheap diagnostics

(implementation 1–2 days; compute negligible)

A0.1

Fix the exp03 duplicate-answer regression. Add an explicit uniqueness assertion at the underlying clause boundary, not just at the smoke-test level.

A0.2

Run the full canonical suite; archive a machine-readable test manifest (script, exit code, pass count, fail count, wall time, commit hash).

A0.3

Commit or archive exp08/, experiments/scratch_cat.metta, and src/chem_catalysis.metta so the source of truth is unambiguous. Mark exp08 explicitly as *not preregistered*.

A0.4

Tag a review baseline commit.

A0.5

New, and do this first. For every existing chemistry (exp04, exp06, exp07 ordered and dual bootstrap), compute the production-route multiplicity histogram $\{r_m\}$. Report it.

A0.6

New. For the same chemistries, estimate the critical seed size $Nx^* = N\phi/(k[\text{resource}])$ from the declared replenishment schedule and dilution, and compare against the maximum abundance any species actually attained in the stored trajectories.

Acceptance. Suite green; manifest archived; two tables produced: (chemistry, $\{r_m\}$) and (chemistry, Nx^* , $\max_x n_x$, verdict). *Expected finding, to be confirmed or refuted:* $r_m \equiv 1$ for the term-based chemistries, and $Nx^* > \max_x n_x$ for several exp07 configurations. Either finding retro-explains a large fraction of the project’s negative results at essentially zero cost.

Phase 1 — Chemistry and RAF oracle

(implementation 3–5 days; compute < 1 CPU-hour)

A1.1

Implement PCAL in PeTTa: associative constructor with AC-matching, mass grading μ , maximum mass M , food set $F = \{\mu \leq m_0\}$, ligation and cleavage families. Assert $r_m = m - 1$ as a test.

A1.2

Implement thinned junction-straddling pattern catalysis with a deterministic seeded hash $h(c, \rho)$. Expose θ as the single density knob. Preserve exact replay.

A1.3

Implement iid Kauffman catalysis behind the *same* interface, parameterised so it can be matched to the pattern model on mean out-degree f . This is the control arm, not a legacy path.

A1.4

Implement **maxRAF** twice: once in a compiled host (Rust or C; needed for the sweep sizes below) and once in PeTTa (needed as the inspectable reference).

A1.5

Implement an exhaustive oracle: for $|R| \leq 18$, enumerate all $2^{|R|}$ subsets and test the RAF predicate directly. $2^{18} = 262\,144$ subsets, trivial.

A1.6

Cross-validate on 500 random small systems spanning: empty food closure; self-catalysis; multiple catalysts per reaction; multiple products; unreachable static cycles; monotonicity when catalysis edges are added; deletion sensitivity.

Acceptance. 500/500 exact agreement between oracle, host, and PeTTa implementations on maxRAF membership (not just size), across all seven edge-case families. Any single disagreement blocks the phase.

Phase 2 — Mean-field predictor, committed in advance

(implementation 1 day; compute minutes)

A2.1

Implement the graded solver of Appendix C: iterate (2)–(3) from $q \equiv 1$ to the greatest fixed point; bisect on f to locate f_c .

A2.2

Emit, for each (k, M, m_0) in the Phase-3 ladder: predicted f_c , γ_c , the full profile $\{q_m\}$, and the mass cutoff $M^* = \max\{m : q_m > 0.5\}$.

A2.3

Commit these predictions to the repository with a hash, in a file explicitly marked prospective, *before* any Phase-3 data is generated.

Acceptance. Predictions file committed and hashed. The solver reproduces the reference values in this document: $f_c = 4.67, 5.70, 6.55, 7.32, 8.04$ for $k = 2$, $m_0 = 1$, $M = 6, 8, 10, 12, 14$; and $r\gamma_c \rightarrow 2.4554$, $q_c \rightarrow 0.7153$ in the large- r scalar limit.

Phase 3 — Structural sweep and threshold calibration

(implementation 1 day; compute 1–4 CPU-hours)

A3.1

Mass ladder $M \in \{6, 8, 10, 12, 14\}$ with $k = 2$, giving $|R| \approx 5.2 \times 10^2, 3.1 \times 10^3, 1.6 \times 10^4, 8.2 \times 10^4, 3.9 \times 10^5$ — just under three decades.

A3.2

25 values of θ , log-spaced so that f spans $[0.3, 3.0] \times f_c^{\text{MF}}$.

A3.3

Replicates: 500, 500, 200, 50, 20 networks per (M, θ) cell respectively.

A3.4

Both catalysis models — thinned pattern and iid — at *matched* mean out-degree f , not at matched θ .

A3.5

Also sweep $m_0 \in \{1, 2\}$ at $M = 12$ to test the food-set prediction (f_c : 7.32 $\rightarrow$ 4.66).

A3.6

Outputs per cell: Pr[RAF exists]; maxRAF size histogram; number of irreducible sub-RAFs (enumerate only for $M \leq 8$); catalytic out-degree histogram with mean, Gini, tail index; empirical γ ; empirical $\{q_m\}$ profile.

Compute sizing. The RAF algorithm converges in tens of outer iterations, each $O(|R| + |X|)$, so $\approx 50|R|$ operations per network. Worst cell ($M = 14$, 20 replicates, 25 θ , 2 models): $50 \times 3.9 \times 10^5 \times 20 \times 25 \times 2 \approx 2 \times 10^{10}$ operations; summed over the ladder, $\approx 4.3 \times 10^{10}$. At the assumed 10^7 effective set-operations per second per core (Appendix B.5), that is **1–4 single-core CPU-hours** for the entire sweep — minutes on a multicore box.

The throughput assumption is load-bearing and depends on representation: hold the closure as a bitset ($|X| = 32766$ bits ≈ 4 KB at $M = 14$, so it stays resident in L1/L2 and the inner loop is cache-local). A pointer-chasing set representation loses one to two orders of magnitude.

This sweep is *not* feasible in PeTTa on SWI-Prolog at these sizes — expect 10^2 to 10^3 times slower, which turns hours into weeks. That gap is the entire reason A1.4 requires two implementations. Use PeTTa to re-verify a random 1% subsample.

Acceptance (four independent checks).

1. The transition *sharpens* monotonically with M (width of the Pr[RAF] curve in f decreases). If it does not, the catalysis is hub-dominated and A3.6's Gini will show it.
2. The maxRAF size histogram is bimodal near threshold, with a visible gap.
3. Measured $f_c \leq$ the Phase-2 prediction at every M , and grows logarithmically in $|R|$.
4. Pattern and iid catalysis give measurably different f_c at matched f , with a reported effect size and confidence interval.

Phase 4 — Dynamical layer

(implementation 3–5 days; compute 5–50 CPU-hours)

A4.1

Implement a Gillespie SSA over PCAL in a CSTR, with tau-leaping for large N . Parameters: N , ϕ , ε , food influx set point.

A4.2

Implement shortest-path enumeration in the uncatalysed-step metric: BFS/Dijkstra on the reachability graph where catalysed edges have weight 0 and uncatalysed edges weight 1. Output the catalytic depth d and the minimal-path multiplicity A for the target closure.

A4.3

Measure R_0 empirically: seed n_0 copies of the core, run to extinction or fixation, estimate R_0 from $P_{\text{surv}} = 1 - R_0^{-n_0}$ across a sweep in n_0 .

A4.4

Measure L_{max} by sweeping the core cycle length L at fixed ϕ and locating the persistence

cutoff.

A4.5

Report all four endpoints (structural / reachable / active / causal) and all five diagnostics separately, per run.

Compute sizing. $N \in \{10^3, 10^4, 10^5\}$, horizon $T = 10^3$ turnover times. SSA event count $\approx NT \approx 10^8$ per trajectory at $N = 10^5$; use tau-leaping above $N = 10^4$ for a 10–30 $\times$ saving. At 100 trajectories per condition and roughly 50 conditions, total event count is $\approx 5 \times 10^{11}$ raw, falling to $\approx 5 \times 10^{10}$ with tau-leaping. At the assumed 3×10^6 events/s/core this is **5–50 single-core CPU-hours**.

That rate assumes a dependency-graph SSA with a priority queue, so that each event updates only the propensities of reactions touching the changed species. A naive $O(|R|)$ propensity recomputation per event at $|R| = 8 \times 10^4$ costs three orders of magnitude more and would make this phase, rather than Phase 6, the dominant compute expense of the program.

Acceptance. Seeded supercritical cores persist and seeded subcritical cores die, with the crossover located within a factor of two of $L_{\max} = \ln A_{\text{br}} / \ln(1/\eta)$. Measured $P_{\text{surv}}(n_0)$ fits $1 - R_0^{-n_0}$ across at least five values of n_0 .

Phase 5 — Rare-event estimators, validated exactly

(implementation 3–5 days; compute minutes)

A5.1

Build the exact CTMC for the tiny chemistry $k = 2$, $M = 3$ ($|X| = 14$, $|R| = 20$) with total population $N \leq 8$. State count $\binom{N+|X|-1}{|X|-1} \approx 2 \times 10^5$ — a trivial sparse system. Compute (i) the exact committor $h(n) = \Pr[\text{ACS before food-only} \mid n]$ by sparse solve, and (ii) the exact emergence probability by matrix exponential or first-passage solve.

A5.2

Implement importance sampling with a per-trajectory likelihood-ratio accumulator, exactly as in (11). Reuse the existing exp07 guidance policies unchanged as proposal distributions.

A5.3

Implement Forward Flux Sampling with progress coordinate $\lambda(n)$ = number of distinct target-closure reactions that have fired at least once in the current excursion. Ladder of 12–20 interfaces, 10^3 trials per interface.

A5.4

Validate A5.2 and A5.3 against A5.1’s exact answer, over at least 10 independent replications.

A5.5

Construct the Doob h -transform from the exact committor and verify the resulting estimator has (numerically) zero variance. This is the check that licenses the vocabulary.

Acceptance. Both estimators recover the exact probability within their reported confidence intervals in at least 9 of 10 replications, at a target probability of 10^{-6} or smaller. The Doob estimator’s variance is below 10^{-12} relative. Only after this passes may “Doob transform” or “Schrödinger bridge” be used in any write-up.

Phase 6 — Test the emergence law

(implementation 2–3 days; compute 50–200 CPU-hours plus pilot tuning)

A6.1

Grid: $N \in \{10^3, 10^4, 10^5\}$, $T \in \{10^2, 10^3, 10^4\}$, $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$, $d \in \{1, 2, 3, 4\}$ by construction of the food set distance. 108 cells.

A6.2

Estimate P (emergence and persistence) in each cell by FFS, with confidence intervals.

A6.3

Fit (8) and test collapse: plot $-\ln(1 - P)$ against $ANk_{\text{cat}}\varepsilon^d(1 - R_0^{-n_0})T$ on log–log axes and check for a unit-slope line.

Compute sizing. 108 cells $\times$ one FFS estimate each. A single estimate at $P \sim 10^{-10}$ costs roughly 15 interfaces $\times 10^3$ trials $\times \sim 3 \times 10^4$ events per short segment $\approx 5 \times 10^8$ events, so the grid is $\approx 5 \times 10^{10}$ events, or **5 CPU-hours** at 3×10^6 events/s/core.

The quoted 50–200 CPU-hours is dominated not by the production grid but by *interface tuning*. Badly spaced ladders ($p_s < 0.05$ or $p_s > 0.6$) waste most of the sampling, and two or three discarded pilot passes per parameter regime is normal. Adaptive Multilevel Splitting trades implementation time for less of this and is worth considering if the pilot cost runs high.

Acceptance. Collapse with $R^2 \geq 0.9$ on $\log P$ across at least 80 of the 108 cells, or a documented and characterised systematic deviation attributed to a named assumption (well-stirredness, single-bottleneck, or seeding/survival independence).

Explicitly out of scope until Phase 3 passes

- Any further candidate-cap, provenance, or attrition-stage instrumentation.
- exp08 in its current 24-permutation form.
- Any new hand-designed pathway chemistry.
- Any guided-versus-unguided incidence comparison.
- Any use of the words “emergence”, “self-organisation”, or “open-ended” in a commit message.

B Consolidated sizing tables**B.1 Structural sweep**

M	$ X $	$ R $	replicates	f_c^{MF}	γ_c target ($2.4/r$, $r = M - 1$)
6	126	516	500	4.67	0.480
8	510	3 076	500	5.70	0.343
10	2 046	16 388	200	6.55	0.267
12	8 190	81 924	50	7.32	0.218
14	32 766	393 220	20	8.04	0.185

25 θ values $\times$ 2 catalysis models per row. Total $\approx 2 \times 10^{10}$ operations; single-digit CPU-hours in a compiled host.

B.2 Direct-simulation reach: required NT

Required NT from (10), with $A = k_{\text{cat}} = 1$. Read down the column for your catalytic contrast δ , across for your bootstrap depth d .

d	$\delta = 0.1$	$\delta = 0.05$	$\delta = 0.01$	feasibility (direct SSA)
1	10	2×10^1	10^2	trivial
2	10^2	4×10^2	10^4	trivial
3	10^3	8×10^3	10^6	easy
4	10^4	1.6×10^5	10^8	feasible
5	10^5	3.2×10^6	10^{10}	borderline
6	10^6	6.4×10^7	10^{12}	needs FFS
8	10^8	2.6×10^{10}	10^{16}	needs FFS

Where exp07 sat. Dual bootstrap: $d = 6$, $N \sim 10^1$, $T = 32$, so $NT \sim 3 \times 10^2$. Required at $\delta = 0.05$: 6.4×10^7 . Shortfall: five orders of magnitude.

Where to sit instead. Design the chemistry for $d \leq 3$, run $N = 10^4$ and $T = 10^3$ ($NT = 10^7$), and you have three to four orders of magnitude of headroom at $\delta = 0.01$. Everything beyond $d = 4$ goes through Phase 5’s estimators rather than through bigger runs.

B.3 Rare-event cost versus direct simulation

target P	direct runs needed	FFS stages ($p_s = 0.2$)	FFS trajectory segments
10^{-4}	10^5	6	6×10^3
10^{-6}	10^7	9	9×10^3
10^{-8}	10^9	12	1.2×10^4
10^{-10}	10^{11}	15	1.5×10^4
10^{-12}	10^{13}	17	1.7×10^4

Segments are short (one interface crossing), so the cost grows *logarithmically* in $1/P$ rather than linearly. This is the single highest-leverage change available to the project.

B.4 Exact-validation system

k (alphabet)	2
M (max mass)	3
$ X $	14
$ R $ (ligations)	20
max total population N	8
CTMC state count	$\approx 2 \times 10^5$
sparse solve	seconds

B.5 Compute accounting

All compute figures in this document are **single-core CPU-hours** on a compiled host (Rust or C), under the throughput assumptions stated below. Implementation time — agent wall-clock spent writing and validating code — is quoted separately in each phase heading and does not parallelise.

Phase	Dominant workload	Throughput assumed	Single-core cost
0	two static diagnostics	—	negligible
1	2^{18} -subset oracle, 500 systems	10^7 set-ops/s	< 1 CPU-hour
2	graded fixed-point solve	—	minutes
3	4.3×10^{10} set-ops	10^7 set-ops/s	1–4 CPU-hours
4	$\sim 5 \times 10^{10}$ SSA events	3×10^6 events/s	5–50 CPU-hours
5	2×10^5 -state sparse solve	—	minutes
6	108 FFS cells + pilot tuning	3×10^6 events/s	50–200 CPU-hours
Total			$\approx 10^2$ CPU-hours

The program is design-limited, not compute-limited. The whole thing is order 10^2 single-core CPU-hours: one workstation over a weekend, or an afternoon on 32 cores. Every workload is embarrassingly parallel at the replicate or grid-cell level with no shared state, so wall time divides by core count essentially exactly. Nothing here needs a cluster, and no result in this document is gated on hardware. This is worth stating plainly because the review packet’s history reads as though more compute were the missing ingredient; it was not, and it still is not.

Three sensitivities that dominate the estimates.

1. **Implementation language.** The 10^7 set-ops/s figure assumes bitset closures resident in L1/L2 cache. PeTTa on SWI-Prolog should be expected to run 10^2 – 10^3 times slower, which moves Phase 3 from hours to weeks and Phase 6 out of reach. Hence the two-implementation requirement in A1.4: compiled host for throughput, PeTTa for inspectability, cross-validated on a 1% subsample.
2. **SSA propensity updates.** The 3×10^6 events/s figure assumes a dependency-graph SSA with a priority queue. Naive $O(|R|)$ recomputation per event at $|R| = 8 \times 10^4$ costs three orders of magnitude and makes Phase 4 the dominant expense.
3. **FFS interface placement.** Most of the Phase 6 range is discarded pilot runs, not the production grid. Keep per-stage crossing probabilities in $[0.05, 0.6]$ and instrument p_s from the first pilot onward.

Memory. Not a constraint anywhere in this program. The largest object is the $M = 14$ reaction index at a few hundred megabytes; the $M = 12$ ladder rung fits comfortably in tens of megabytes, and the exact CTMC of B.4 is a sparse matrix of order 10^5 rows.

C Reference implementations

C.1 The RAF algorithm (greatest fixed point)

```
function maxRAF(R, F, catalysis, reactants, product):
  S := R
  repeat
    # closure of food under S
    C := F
    changed := true
    while changed:
      changed := false
      for rho in S:
        if reactants(rho) subset-of C and product(rho) not-in C:
          C := C + {product(rho)}
```

```

        changed := true
    # prune
    S' := { rho in S :
        reactants(rho) subset-of C
        and exists c in C with c catalyses rho }
    if S' == S: return (S, C)
    S := S'
until S is empty
return (empty, F)

```

Cost: outer loop is monotone decreasing, so at most $|R|$ iterations, in practice tens. Each iteration is $O(|R| + |X|)$ with the right indices. Budget $50|R|$ operations per network.

C.2 Graded mean-field solver

```

function meanfield(k, M, m0, f):
    R := sum over m=2..M of (m-1) * k^m
    pi := f / R
    q := array of ones, index 1..M
    repeat until fixed point (max |dq| < 1e-14):
        C := sum over m=1..M of (k^m * q[m])
        gamma := 1 - exp(-pi * C)
        for m in m0+1 .. M:
            prod := 1
            for i in 1 .. m-1:
                prod := prod * (1 - q[i]*q[m-i]*gamma)
            qnew[m] := 1 - prod
        for m in 1 .. m0: qnew[m] := 1
        q := qnew
    return q, gamma

function critical_f(k, M, m0):
    # bisection in log space on the largest f with q[M] ~ 0
    lo, hi := 1e-6, 1e6
    repeat 60 times:
        mid := sqrt(lo*hi)
        if meanfield(k,M,m0,mid).q[M] > 1e-6: hi := mid
        else: lo := mid
    return hi

```

Initialising at $q \equiv 1$ and iterating downward is essential: it converges to the *greatest* fixed point, mirroring the RAF algorithm. Initialising at $q \equiv 0$ converges to the trivial fixed point and tells you nothing, because the transition is a saddle-node.

C.3 Importance-sampling weight accumulator

```

# run one trajectory under tilted propensities atilde,
# accumulating the exact likelihood ratio w = dP/dQ

w := 1.0
t := 0
n := initial_state
while t < T:
    a := true_propensities(n) # vector over reactions
    at := tilted_propensities(n) # vector over reactions
    a0, at0 := sum(a), sum(at)
    tau := Exponential(rate = at0)

```



```

    if t + tau > T:
        w := w * exp((at0 - a0) * (T - t))
        break
    rho := sample_index(probabilities = at / at0)
    w := w * (a[rho] / at[rho]) * exp((at0 - a0) * tau)
    n := apply(rho, n)
    t := t + tau

# unbiased estimate of E_P[Psi]
contribution := Psi(trajecory) * w

```

Report both the estimate and the effective sample size $ESS = (\sum w_i)^2 / \sum w_i^2$. An ESS below about 50 means the tilt is bad and FFS should be used instead.

C.4 Forward Flux Sampling

```

# lambda(n) = number of distinct target-closure reactions
# fired at least once in the current excursion

interfaces := [L0 < L1 < ... < LS]
P := 1
configs := collect_crossings_of(L1, from = basin_at_L0, count = K)

for s in 1 .. S-1:
    successes := 0
    new_configs := []
    for trial in 1 .. Ntrials:
        n := uniform_choice(configs)
        run SSA from n until lambda >= L(s+1) (success)
            or lambda <= L0 (failure)

        if success:
            successes := successes + 1
            new_configs.append(current_state)
    p_s := successes / Ntrials
    P := P * p_s
    configs := new_configs
    if p_s < 0.05: warn "interface spacing too wide"
    if p_s > 0.6: warn "interface spacing too narrow"

return P, per-stage p_s with binomial CIs

```

D Parameter glossary

Symbol	Name	Meaning and where it is set
k	alphabet size	Number of atoms. Design choice; 2 or 4.
M	maximum mass	Longest molecule. Sets $ X \approx k^M$, $ R \approx (M - 1)k^M$.
m_0	food mass	All molecules of mass $\leq m_0$ held at influx. Strong lever on f_c .
$\mu(s)$	mass	Word length; conserved by ligation/cleavage.
r_m	route multiplicity	Number of reactions producing a given mass- m molecule. $m - 1$ for associative, 1 for free terms. Must exceed 1.

Symbol	Name	Meaning and where it is set
$\sigma(c)$	specificity	Non-wildcard positions in catalyst c . Controls out-degree exponentially.
θ	thinning	Catalysis acceptance rate. The single sweep knob; $f = \theta f_{\max}$.
π	pair catalysis prob.	$f/ R $.
f	mean out-degree	Expected reactions catalysed per molecule. Threshold at $f_c \approx 5-8$.
γ	catalysed fraction	Probability a reaction has an available catalyst. Threshold $\gamma_c \approx 2.4/r$.
q_m	closure fraction	Probability a random mass- m molecule is in the maxRAF closure.
M^*	mass cutoff	Largest m with $q_m > 1/2$. Predicted structural signature.
N	population	Total molecule count in the chemostat.
T	horizon	Run length in turnover times.
ϕ	dilution rate	Chemostat outflow. Sets η and x^* .
ε	catalytic contrast	k_0/k_{cat} , uncatalysed over catalysed rate.
δ	contrast threshold	Largest ε at which catalysis is still identifiable.
d	catalytic depth	Minimum uncatalysed steps to bootstrap the target closure. Design to ≤ 3.
A	path multiplicity	Number of distinct minimal bootstrap paths. Measured by shortest-path enumeration.
L	core cycle length	Steps in the autocatalytic core. Bounded by $L_{\max}$.
η	step efficiency	$k_{\text{cat}}/(k_{\text{cat}} + \phi)$.
R_0	reproductive number	$A_{\text{br}}\eta^L$. Must exceed 1 for persistence.
n_0	seed size	Initial copy number of the core.
x^*	critical seed	$\phi/(k[\text{resource}])$; basin boundary. Check $Nx^* \leq \max_x n_x$.